

IOANNIS KALFAS

Senior AI Engineer | GeoAI, Computer Vision, Edge AI and Production ML | Greek, EU national

kalfasyan@pm.me · (+32) 465 40 30 27 · Leuven, Belgium · [GitHub](#) · [LinkedIn](#) · [Google Scholar](#)

PROFESSIONAL SUMMARY

Senior AI Engineer and PhD with 11 years of experience building computer-vision and GeoAI systems from research prototype to operational deployment. Combines deep-learning model development with data engineering, reproducible MLOps, and optimization for cloud, edge, and embedded environments. At VITO, developed and operationalized global land-cover and cloud-screening workflows for LCFM before transitioning to the GeoAI team to focus on efficient models for resource-constrained edge and onboard computing. Known for connecting scientific modeling, maintainable software, and production infrastructure across multidisciplinary teams.

Highlights: *global-scale LCFM deployment for Copernicus · 9 peer-reviewed publications · open-source tools in active research use*

WORK EXPERIENCE

R&D Engineer | VITO | Mol

Mar 2024 – Present

GeoAI and Edge Solutions

Jan 2026 – Present

- Optimize and evaluate CORSA VQ-VAE architectures for satellite-image compression, balancing model footprint with spatial and spectral reconstruction quality on resource-constrained edge and embedded platforms.
- Develop internal MLOps tools for experiment tracking, reproducible configuration, and job management on VITO's on-premises GPU cluster.
- Build parametrized training workflows using PyTorch, MLflow, Hydra, and Pydantic to improve experiment reproducibility and model traceability.

Remote Sensing Applications (ESA WorldCover & Copernicus LCFM team)

Mar 2024 – Jan 2026

- Developed deep-learning models for cloud segmentation and *global* 10 m land-cover mapping within the Copernicus Land Cover and Forest Monitoring programme.
- Built multi-stage pipelines for satellite-data ingestion, preprocessing, model inference, and generation of annual global Earth-observation products.
- Implemented reusable cloud-model inference workflows across local, AWS, and openEO environments.
- Engineered distributed data pipelines using Hadoop, Spark, object storage, and network storage for geographically and seasonally diverse satellite datasets.
- Improved model operationalization through MLflow integration, package maintenance, output validation, and standardized GeoTIFF metadata.
- Supported technical handover by guiding colleagues through model configurations, datasets, annotation sources, training code, and inference workflows.

Postdoctoral Researcher → PhD Researcher | KU Leuven, MeBioS | Leuven Dec 2018 – Mar 2024

Postdoctoral Researcher

Nov 2022 – Mar 2024

- Led the adoption of shared Git infrastructure and software-engineering practices across multiple research teams.
- Organized bimonthly deep-learning meetups for researchers across departments, communicating technical concepts to interdisciplinary audiences.
- Managed multi-terabyte research datasets through KU Leuven's ManGO platform, improving dataset organization and accessibility.
- Built and deployed an AWS-hosted image-classification, annotation, and model-serving platform used by research teams and external collaborators.
- Investigated hyperspectral imaging and machine-learning methods for early disease detection in agrifood applications.

PhD Researcher in Pest-Insect Identification

Dec 2018 – Nov 2022

- Designed multimodal datasets and training pipelines for optical and acoustic insect identification using image and time-series data.
- Developed deep-learning models for Raspberry Pi devices and integrated them with an AWS-hosted API serving external research collaborators.
- Supervised more than eight master's and bachelor's thesis students in computer vision and signal processing.
- Published five peer-reviewed papers on AI-driven biological monitoring.

Data Scientist / AI Engineer | [Faktion](#) | Antwerp

Dec 2017 – Dec 2018

- ▶ Developed end-to-end machine-learning solutions for industrial computer vision, speech processing, and predictive-maintenance use cases.
- ▶ Standardized feature-extraction and data-preparation workflows for heterogeneous client datasets, improving consistency and reuse across projects.
- ▶ Won an industry hackathon by developing a video-based activity-recognition prototype.

Doctoral Researcher in Computational Neuroscience | [KU Leuven, Neurophysiology](#) | Leuven

Nov 2015 – Nov 2017

- ▶ Investigated deep convolutional networks as computational models of biological visual processing.
- ▶ Developed computer-vision and regression models to predict neuronal responses to visual stimuli.
- ▶ Published four peer-reviewed papers, including two as first author, and presented at the Vision Sciences Society conference before discontinuing the programme to pursue applied AI in industry.

TECHNICAL SKILLS

GeoAI and Computer Vision: Satellite imagery, semantic segmentation, image classification, cloud screening, land-cover mapping, multispectral and hyperspectral imaging

Edge and Efficient AI: VQ-VAE, neural image compression, model optimization, constrained inference, Raspberry Pi, embedded deployment

ML Engineering and MLOps: PyTorch, PyTorch Lightning, MLflow, Hydra, Pydantic, Scikit-learn, experiment tracking, model validation

Data Engineering: Spark, Hadoop, Pandas, xarray, GeoTIFF, distributed processing, large-scale ETL

Cloud and Infrastructure: AWS EC2/S3, Docker, Linux, Bash, FastAPI, REST APIs, on-premises GPU clusters

Data Applications: Streamlit, NiceGUI, Solara

Languages: Greek (native), English (professional), Dutch / Spanish / Swedish (elementary)

EDUCATION

PhD, Bioscience Engineering — [KU Leuven](#), Belgium 2022

MSc, Machine Learning — [KTH Royal Institute of Technology](#), Stockholm, Sweden 2015

BSc, Computer Science — [Aristotle University of Thessaloniki \(AUTH\)](#), Greece 2013

RESEARCH AND OPEN-SOURCE CONTRIBUTIONS

- ▶ Author or co-author of nine peer-reviewed publications across computer vision, bioscience AI, and computational neuroscience, including two first-author neuroscience papers.
- ▶ Develop and maintain open-source Python tools including desto for job management, vresto for Sentinel-data discovery, plakakia for image tiling, and filoma for research-data organization.